

Chapter 1 Introduction

1.1 Overview

1.1.1 Climate, weather

Climate and weather are two closely interrelated yet distinct concepts in atmospheric science. Weather refers to day to day state of the atmosphere which changes quickly while climate is the average weather pattern in a place over a long period of time. In recent decades, the intensity of extreme weather events marked an increase, raising concerns among scientists, policymakers and general public. These results in devastating socio-economic and environmental consequences.

1.1.2 Definition of Extreme Events

When scientists use the term extreme climatic conditions, they are referring to the weather patterns that are far outside the normal range for human, animal and plant life. Extreme climate refers to the regions of the world where the average weather pattern is quite harsh and living beings in such conditions need to adapt special survival features. While extreme weather events refer to places with a normal climate experiencing a sudden, severe weather event that breaks the historical records.

1.1.3 Statistical Definition

Since extreme weather events are commonly understood as rare occurrences, these events represent significant deviations from the usual climate patterns in probability distribution of a given climate variable, such as temperature and precipitation. Or in other words, an event is considered extreme when it crosses a predefined threshold either in terms of intensity, frequency or duration and as high potential to for adverse impacts on human societies and natural ecosystems.

1.1.4 WMO Definition

The World Meteorological Organization (WMO) provides the most widely accepted international framework for defining and monitoring climate extremes. According to WMO guidelines, these events are often identified using statistical thresholds relative to the local climatological baseline, typically spanning a 30-year reference period. The WMO along with the ETCCDI (Expert Team on Climate Change Detection and Indices) have created 27 standard measurements for tracking extreme weather for consistent monitoring of extremes across the globe. These measurements use two approaches – i) percentile-based indices, ii) threshold-based indices that allows insightful comparison across different climatic regions. This framework serves as the foundation for evaluating the performance of climate models such as CMIP6 ensemble in simulating historical and future extreme climatic conditions.

1.1.5 Climate Extreme around the globe

On a global level, extreme weather events have become more common and stronger in many parts of the world. Due to climate change, events like heat waves, droughts and heavy rainfall are increasing. Scientists believe that as the Earth gets warmer, the atmosphere's moisture holding capacity increases approximately by 7% for every 1°C rise in temperature, thereby enhancing the potential for extreme precipitation events. Many countries have faced serious problems because of extreme rainfall. For example, massive floods have occurred in Europe, Pakistan and different parts of India. In India, the monsoon season often brings very heavy rainfall in a short period of time causing flash floods, riverine flooding and landslides. These global and regional trends highlight the urgent need to improve our understanding and predictive capabilities of extreme weather events. As climate model evolved from CMIP5 to the more advanced CMIP6 generation, it has become more important to evaluate how well these models can simulate different types of climates extremes like heat waves, cold waves and heavy rainfall.

1.1.6 Societal awareness and public understanding

Extreme weather events do not only affect the environment but also the society. How people understand and respond to these events depends on their awareness and knowledge about climate change. In many parts of the world, there is a growing need to improve how the information about extreme weather is shared with the general public.

Media plays an important role in spreading the information. However, the news sometimes focuses only on damages which creates fear rather than giving useful guidance. Risk communication is the process of explaining the danger of extreme events in a simple and clear way so that people can timely take the actions.

Weather agencies like India Meteorological Department also plays a vital role by timely issuing warnings for heavy rainfall, heat waves or cold waves. Government advisories and disaster management systems help in spreading alerts through radio, mobile etc.

Improving public understanding and awareness is essential for building a climate-resilient society. Better education and effective warning systems can help people and governments take actions timely against extreme weather events.